QUIZ 1 - The Baku Toki Queue

You manage “Baku Toki" attraction in theme park which uses electric Bumper Car. Each car can accommodate 1-2 visitors. You sells 2 types of tickets, namely FAST_TRACK and REGULER. The FAST_TRACK ticket allows visitors to skip the REGULER queue and get priority.
To accommodate the process, you are given a challenge **a Queue management application using C programming and a doubly linked list.**

Your application should be able to:

- **Register visitor** with name, and FAST_TRACK status.
    - **Sort tasks** automatically : FAST_TRACK visitors queued (First in Fist Out) ahead of REGULAR visitors
- **Calling** 1-2 visitors to drive a car and display the name(s)
- **Cancel Visitor** queue by name and display the result.
- **Display the current list of visitor after registration**

Task Requirements
Your program should follow these specifications:

1. INPUT FORMAT

- The first line of input contains two integers:

  N M

    - N → Number of visitors to be added.

- M → Number of operations to be performed on the list.

- The next N lines contain:

  FAST_TRACK NAME

  - FAST_TRACK → Integer representing the FAST_TRACK status (1 is FAST_TRACK, 0 is REGULAR).
  - NAME → String representing the visitor'name. Name can have more than one word. Maximum 99 characters.
  - Each visitor data should be stored in a **doubly linked list**, **sorted by FAST_TRACK status descending.**

- After inserting all tasks, the next M lines represent operations, which can be:

  OPERATION [NAME]

  - **CALL** → Removes 1-2 person in front of the queue depends on the how many people available in the queue. If the last person in the queue is only 1 person. The last person will drive the car him/herself.
  - **CANCEL NAME** → Removes the **visitor with specific name**, if it exists.

2. Expected Output

- After performing all operations, the program should print all **visitors list in the queue** in the following format:
  Queue# Name FAST_TRACKStatus
  e.g. :
  1 Max Verstappen 1
  2 Lando Norris 1
  3 Charles Leclerc 1
  4 Oscar Piastri 1
  5 Carlos Sainz 1
  6 George Russel 0
  7 Lewis Hamilton 0
  8 Sergio Perez 0
  9 Fernando Alonso 0

- After performing **CALL** operation, the program should print the name of visitor inside each car.
  Max Verstappen – Lando Norris
  Charles Leclerc – Oscar Piastri
  Carlos Sainz – George Russel
  Lewis Hamilton – Sergio Perez
  Fernando Alonso

- After performing **CANCEL** operation,
    - If an operation tries to remove a non-existent visitor, print '**NAME** is not exist'. e.g. : 'Michael Schumacher is not exist'
    - If the name exist, print '**NAME** is removed' e.g. : 'Fernando Alonso is removed'

Example Input & Expected Output

Case 1 :

Input	Output
9 5 1 Max Verstappen 1 Lando Norris 0 George Russel 0 Lewis Hamilton 1 Charles Leclerc 1 Oscar Piastri 0 Sergio Perez 1 Carlos Sainz 0 Fernando Alonso CALL CALL CALL CALL CALL	1 Max Verstappen 1 2 Lando Norris 1 3 Charles Leclerc 1 4 Oscar Piastri 1 5 Carlos Sainz 1 6 George Russel 0 7 Lewis Hamilton 0 8 Sergio Perez 0 9 Fernando Alonso 0 Max Verstappen – Lando Norris Charles Leclerc – Oscar Piastri Carlos Sainz – George Russel Lewis Hamilton – Sergio Perez Fernando Alonso

explanation:

- Visitors' data inserted into the list are **sorted by FAST_TRACK status descending, First in First Out (FIFO)**
- Display the data in the LL.
- Display each Operation Output : 5 x print the name of visitor inside each car.

Case 2 :

Input	Output
9 8 1 Max Verstappen 1 Lando Norris 0 George Russel 0 Lewis Hamilton 1 Charles Leclerc 1 Oscar Piastri 0 Sergio Perez 1 Carlos Sainz 0 Fernando Alonso CANCEL Max Verstappen	1 Max Verstappen 1 2 Lando Norris 1 3 Charles Leclerc 1 4 Oscar Piastri 1 5 Carlos Sainz 1 6 George Russel 0 7 Lewis Hamilton 0 8 Sergio Perez 0 9 Fernando Alonso 0 Max Verstappen is removed Carlos Sainz is removed

CANCEL Carlos Sainz CANCEL George Russel CANCEL Fernando Alonso CANCEL Michael Schumacher CALL CALL CALL	George Russel is removed Fernando Alonso is removed Michael Schumacher is not exist Lando Norris - Charles Leclerc Oscar Piastri - Lewis Hamilton Sergio Perez

explanation:

- Visitors' data inserted into the list are **sorted by FAST_TRACK status descending, First in First Out (FIFO)**
- Display the data in the LL.
- Display each Operation Output :
    - Cancel each exist visitor, display the output accordingly if the visitor is exists or not.
    - 3 x print the name of visitor inside each car.

Submit your code with this file format: **Class_NIM_NAMA.CPP**
(Ex: LD07_2801234567_NAME.CPP)

Happy coding!